Progress Report 2: Encoder Bandwidth, Not Foveation Specifically, Shapes Spatial Memory in PointNav Agents

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I. Research question and headline finding

Question. A PointNav agent [1], [2] has two routes to the spatial information its policy needs to act: recompute it each step from the current visual encoder output, or maintain it across time inside the recurrent (LSTM) memory. These routes compete for finite hidden-state capacity. *Does the structure of the visual input determine how this capacity gets allocated?*

Headline finding. Across five conditions that share architecture, sensor stack, training budget, and within-episode behaviour (96–99% success), the linear top-layer GPS R^2 falls *monotonically* with encoder spatial output dimension: from blind ($\approx +0.79$, no encoder) and coarse ($+0.58$, 1×1), through foveated-logpolar (-0.12, 2×2), to foveated-blur ($+0.16$, 4×4) and uniform (-1.19, 4×4). The same hidden states recover GPS at $R^2 \in [0.46, 0.92]$ under a 2-layer MLP probe, so the contrast is about *linear* top-layer readability (the operation the DD-PPO action head performs), not information loss. **Theoretical claim:** bottleneck conditions force the LSTM to integrate the GPS sensor across time and store the result in a linearly-readable axis of $\mathbf{h}_2$; rich-encoder conditions reallocate this capacity to a per-step visual-feature route.

II. Five conditions on one bandwidth axis

All agents share a 3-layer LSTM (512-d) and the Wijmans [3] non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar); only the visual input changes. All four sighted conditions were retrained from scratch under unified hyperparameters (seed=0, num_envs=16, 250M frames), eliminating the hyperparameter inconsistency present in our *Report 1* runs. All achieve 96–99% success and SPL $\in [0.85, 0.89]$, so any cross-condition representational difference cannot be attributed to a skill gap.

Table I

Five conditions, ordered by encoder spatial output. Blind uses an independently-retrained checkpoint at 340M frames.

Condition	**Encoder out**	**Visual input**
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\max}$ = 8)
Uniform	4×4	256×256 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes ($\sim$ 30k–120k LSTM steps) and run six analytical lenses on the top-layer hidden state $\mathbf{h}_2$: linear (Ridge α=10) and MLP (256 hidden, L_2=10^{-4}) probes for GPS; lag-k probe for $k \in \{0, 1, 2, 5, 10, 20, 50\}$; per-(unit, scene) Skaggs spatial information (rectified $\max(\mathbf{h}_2, 0)$); cross-training probes at five intermediate checkpoints; cross-condition CKA, Procrustes shape distance, and memory transplant. All probes use 5-fold episode-level cross-validation. Single seed per condition follows the comparative-cognition modelling convention; rigour at the cross-condition level instead derives from convergence across the six lenses.

III. Results: a Because-But-Therefore chain

Because: linear GPS R^2 falls monotonically with encoder bandwidth. Fitting Ridge on $\mathbf{h}_2 \to$ GPS gives the headline cross-condition pattern (Table II, columns 2–3): bottleneck conditions retain the linear code (blind $+0.79$, coarse $+0.58$); rich-encoder conditions lose it (foveated $+0.16$, uniform -1.19). The third rich-encoder condition (foveated-logpolar, 2×2) sits at -0.12, closer to the rich-encoder regime than to coarse. Distance-to-goal, a same-dimensional control target, decodes at $R^2 \in [0.81, 0.96]$ in every condition, ruling out a probe-capacity-collapse reading.

But: a non-linear (MLP) probe recovers GPS in every condition. A 2-layer MLP fit to the same hidden states recovers GPS at $R^2 \in [0.46, 0.92]$. The MLP-minus-linear gap widens monotonically with encoder bandwidth ($+0.13$ blind $\to$ $+0.22$ coarse $\to$ $+0.59$ foveated-logpolar $\to$ $+0.60$ foveated $\to$ $+1.65$ uniform). *Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions that the DD-PPO linear action head cannot read.*

Table II

Linear / MLP probes for GPS plus lag-k stability summary. (Blind†: preliminary 100-ep partial NPZ from in-flight 500-ep collection.)

	Linear R^2	MLP R^2	MLP–Lin gap	Lag-k (k=0, 20)
Blind†	$+0.79$	$+0.92$	$+0.13$	$+0.78$, $+0.68$
Coarse	$+0.58$	$+0.79$	$+0.22$	$+0.58$, $+0.53$
Fov-logpolar	-0.12	$+0.47$	$+0.59$	-0.03, -0.12
Foveated	$+0.16$	$+0.76$	$+0.60$	$+0.18$, -0.07
Uniform	-1.19	$+0.46$	$+1.65$	-1.78, -6.53

Therefore: capacity allocation, not information loss. The bottleneck (no encoder, or 1×1) forces the LSTM to integrate the GPS sensor across time — it has no other route. A rich encoder offers a viable per-step visual-feature route to the same information; policy gradient reallocates capacity to the visual route, and the integrated code is no longer the linearly-readable axis. *The same hidden state encodes position in both regimes; only the format differs.*

IV. Convergent evidence

Three additional lenses corroborate the capacity-allocation account.

Substitution dynamics across training. Re-probing intermediate checkpoints ($\approx 50/100/150/200/250$M frames) shows that *all four sighted conditions start at high linear GPS R^2 at $\sim 50M$ frames* (coarse $+0.91$, foveated $+0.91$, uniform $+0.83$, foveated-logpolar $+0.92$); the linear top-layer code emerges early across the bandwidth spectrum. Bottleneck (coarse) preserves R^2 around $+0.5$ through training; rich-encoder conditions decay — uniform crosses zero by $\sim$ 200M and lands at -1.79; foveated drifts to $+0.18$; foveated-logpolar to -0.03. Decay rate tracks encoder informativeness about position — this is capacity allocation as a *training-time process*, not a property of the converged endpoint.

Lag-k temporal stability. Decoding past position GPS_{t-k} from current $\mathbf{h}_t$ across $k \in \{0, 1, 2, 5, 10, 20\}$ separates *integrated* from *reactive* memory (Table II, column 5). Bottleneck conditions sustain stable positive R^2 across k (blind: $+0.78 \to +0.68$; coarse: $+0.58 \to +0.53$); rich-encoder conditions are unstable (foveated: $+0.18 \to -0.07$; uniform sub-baseline at k=0 already, crashing to -6.53 at k=20).

Place-cell signature is condition-flat. Per-(unit, scene) Skaggs spatial information (rectified $\max(\mathbf{h}_2, 0)$) is essentially condition-flat: 1.20 (coarse), 1.24 (foveated), 1.19 (uniform), 1.16 (foveated-logpolar); blind 1.35 (preliminary). $\Delta \approx 0.15$ bits, well under the within-condition spread. *The bandwidth-allocation signal lives in the linear-readout axis, not the per-unit place-cell metric.*

V. What changed since Report 1; status; risks

Reframe. *Report 1* reported foveated as having the longest temporal memory (R^2=0.629 at lag-5, the only condition still positive). *This is overturned.* The Report 1 ranking reflected partially-converged training plus heterogeneous hyperparameters across conditions; after unified retraining, foveated's lag-k is *less* stable than coarse's, not more. The new mechanism (encoder bandwidth) makes the same predictions for blind/coarse/uniform/foveated and additionally predicts that foveated-logpolar (a $2{\times}2$ rich-encoder control not in Report 1) lands in the rich-encoder regime — which it does (-0.12).

Status. Phase A probe-collect: $4/5$ complete; blind 500-ep run in flight on EPFL RCP ($\sim$ 6h ETA, partial 100-ep NPZ already used above). Phase B (linear/MLP/lag-k/Skaggs/substitution/CKA-Procrustes-transplant) complete for the four sighted conditions; the blind row of each is queued to fire automatically once the 500-ep NPZ lands. Pre-DDL queue: re-run lenses with converged blind, apply values to the NeurIPS submission, sanity-rebuild the paper PDF.

Risks tracked. (a) The 50-ep blind preview gave catastrophically wide CV folds (linear -0.27 ± 1.41); at 100 eps the same probe stabilised at $+0.79 \pm 0.25$. We expect 500 eps to tighten std to ~ 0.10. (b) A separately-trained blind on a different seed/effective-batch gives a different mean ($+0.36$); we use our own retrained checkpoint and treat the alternative as one cross-setup robustness check.

Contributions. W. Xu post-retrain probe-collect, six-lens analysis pipeline, NeurIPS paper revision, this report. **L. Bruneau** foveation transform, training infrastructure, learned-gaze decoder, CKA/Procrustes pipelines. **Z. Aoutir** substitution-dynamics figure, place-cell visualisations. **R. Abkari** behavioural ablations (shortcut, GPS-zeroing, excursion-forgetting), MP3D evaluation.

References

[1] E. Wijmans, A. Kadian, A. Morcos, S. Lee, I. Essa, D. Parikh, D. Batra, and M. Savva, "DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames," in *International Conference on Learning Representations (ICLR)*, 2020.

[2] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Batra, and D. Parikh, "Habitat: A platform for embodied AI research," in *International Conference on Computer Vision (ICCV)*, 2019.

[3] E. Wijmans, M. Savva, I. Essa, S. Lee, A. Morcos, and D. Batra, "Emergence of maps in the memories of blind navigation agents," in *International Conference on Learning Representations (ICLR)*, 2023.